


Graphic by RAM

THE COMEBACK TEAM: Nair (centre) with ISRO scientists

the cause, the committee then went into finding which part in the strap-on motor had failed.

After extensive simulations they found it to be a minuscule defect in one of the propellants' regulator to the thrust chamber. Instead of a 16-mm diameter orifice, the regulator's manufacturer had machined it to 17 mm—just a hair thickness more. That error was enough to cause an imbalanced flow of the propellants leading to enormous pressure in the gas generators and excessively high temperatures. This cascaded into a structural failure of the entire engine resulting in a loss of thrust that proved fatal to the vehicle. "Launch vehicles are by nature temperamental beasts and a simple lapse can lead to catastrophic failure," says B.N. Suresh, director of the Vikram Sarabhai Space Centre (VSSC), which is responsible for all rocketry.

When the committee's report was released to the press, Nair was asked who was responsible for the failure. The ISRO chairman said: "I am." Rather than pointing an accusing finger, the team sat down to pin-point the procedures that went wrong at the manufacturing stage and how to avoid them in future. Suresh says they created a 100 per cent check list identifying people who would be responsible. They also created quality assessment teams for each of the critical sub-system teams to detect the minutest of errors. A year and a month later, they were ready to test GSLV-F04—the short span for recovery showing how confident ISRO was of its groundwork.

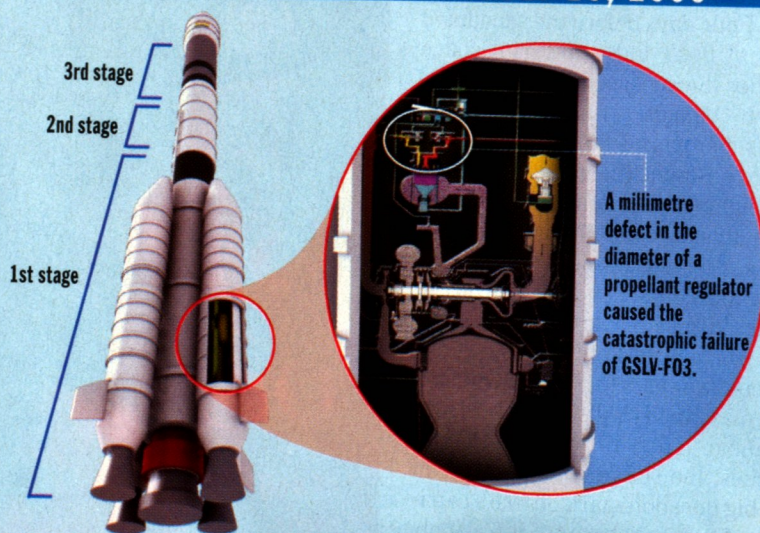
Yet it was not just bad weather that

would result in heart-stopping moments for the team. Just 15 seconds before the lift-off, the computer detected a glitch and put the launch on hold. The team groaned and went into a huddle to detect what had gone wrong. They discovered that a valve in the third stage cryogenic engine which had to close within three seconds, had taken an extra second to do so. The scientists were unsure whether it was a data error or a malfunctioning valve. Nair and top members of his team personally went to oversee the simulation tests. When they were sure it was only a data error, an hour later Nair gave the go-ahead.

With time running out for the

launch window available that day, scientists sat hunched over their consoles or watching the giant screens in the master control room as the countdown reached zero. First, a brilliant flash of light lit the evening sky and then the rocket, with a long tail of fire, rose majestically. As it disappeared into the firmament, there were 17 anxious minutes before it was confirmed that it had successfully ejected INSAT-4CR into the orbit. Bristling with 12 Ku Band transponders, the 2.13 tonne satellite is all set to be the backbone of DTH telecast in the country. Coming after the success of the Space Capsule Recovery Experiment in January, ISRO is back in business, with a bang. ■

WHY GSLV-F02 FAILED ON JULY 10, 2006



THE CAUSE: A hair-thin defect in machining in one of the regulators of a first stage strap-on motor led to imbalanced flow of propellants.

THE EFFECT: A sudden loss of thrust in the strap-on motor resulted in a lack of control and the vehicle broke up because of the excessive stress and heat.